

From Detection to Prediction: A Comparative Assessment of 16 Models for Structural Break, Peak, and Valley Forecasting in Agricultural Commodity Prices

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Abstract

Agricultural commodity price instability poses significant challenges for producers, traders, and policymakers. While numerous statistical methods exist for detecting structural breaks and turning points *ex post*, their capacity for prospective prediction—anticipating events before they occur—remains poorly understood. This paper addresses this gap by conducting a comprehensive walk-forward backtesting comparison of 16 models across three methodological tiers, evaluated on daily price data for eight agricultural commodities (Cattle, Cocoa, Coffee, Corn, Cotton, Soybean, Sugar, Wheat) from 2019 to 2026. The models span classical changepoint methods (Bai-Perron, PELT, Binary Segmentation, CUSUM/ICSS, BCP, E-Divisive, WBS, BOCPD), regime-switching and extreme value approaches (Markov Switching, HMM, PLR+GBM, EVT), and emerging techniques (TDA persistent homology, EEMD+LSTM, Conformal Prediction, Matrix Profile/FLUSS). Our walk-forward framework employs rolling windows of 504 days with 21-day steps, producing 61 evaluation windows per commodity. Key findings reveal that: (i) 9 of 16 models, including all classical changepoint methods except WBS, produce zero predictive signals in the walk-forward setting, demonstrating that retrospective detection capability does not imply predictive power; (ii) Extreme Value Theory achieves the best precision-recall balance for valley prediction ($F1 = 0.28$) with 15-day lead times; (iii) Conformal Prediction leads for peak detection ($F1 = 0.28$) with high recall but elevated false alarm rates; and (iv) no model exceeds $F1 = 0.50$, confirming that prospective event prediction in commodity markets remains fundamentally challenging. These results provide actionable guidance for early warning system design and highlight the critical distinction between detection and prediction in applied time series analysis.

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1. Introduction

Agricultural commodity prices are inherently volatile, driven by supply-side shocks (weather, disease, geopolitical disruptions), demand fluctuations, speculative behavior, and macroeconomic conditions (Gilbert, 2010; Tadasse et al., 2014). The ability to anticipate structural breaks, price spikes, and turning points has direct implications for food security policy, supply chain risk management, and financial hedging strategies (Jacks and Stuermer, 2020). Recent events—the COVID-19 pandemic in 2020 and the Russia-Ukraine conflict in 2022—have underscored how rapidly and severely commodity markets can be disrupted (Amin et al., 2024; Hasanov et al., 2024).

A rich econometric literature addresses the *detection* of structural breaks and regime changes in commodity price series. Classical methods such as Bai-Perron testing (Bai and Perron, 2003), PELT (Killick et al., 2012), and Binary Segmentation (Fryzlewicz, 2014) are well-established for identifying changepoints in retrospect. Regime-switching models (Luo et al., 2022; de la Torre-Torres et al., 2020) and extreme value theory (van Oordt et al., 2021) have been applied to characterize price dynamics and tail risks. More recently, machine learning approaches (Brignoli et al., 2024; Sari et al., 2024), signal decomposition methods (Wu et al., 2019; Zhou et al., 2022), and topological data analysis (Guritanu et al., 2025; Baitinger and Flegel, 2021) have emerged as promising tools for financial time series analysis.

Despite this methodological diversity, a critical gap persists in the literature: **the distinction between retrospective detection and prospective prediction is rarely tested rigorously.** Most comparative studies evaluate models on full-sample data, where the model has access to the entire time series, including observations after the event of interest. This creates a fundamental look-ahead bias that inflates apparent performance. The few studies that employ out-of-sample evaluation (Hasanov et al., 2024; Feng et al., 2023) typically focus on a single method class or a narrow set of commodities.

This paper makes three contributions to the literature:

1. **Comprehensive model comparison under genuine prediction conditions.** We evaluate 16 models spanning three methodological tiers—classical changepoint methods, regime-switching/extreme value approaches, and emerging ML/topological techniques—using a strict walk-forward backtesting protocol that prohibits information leakage.
2. **Explicit separation of detection from prediction.** By contrasting full-sample detection performance with walk-forward predictive accuracy, we quantify the gap between what models can identify retrospectively

and what they can anticipate prospectively. This distinction, while conceptually obvious, has not been systematically documented across a broad model universe.

3. **Multi-commodity, multi-event evaluation.** We assess predictive performance for three event types (structural breaks, peaks, and valleys) across eight agricultural commodities, providing a granular view of where prediction succeeds and fails, and which commodity-event combinations are more amenable to early warning.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data, models, and walk-forward evaluation framework. Section 4 presents the results. Section 5 discusses implications and limitations. Section 6 concludes.

2. Literature Review

2.1. Structural Break Detection

The foundational framework for structural break detection in regression settings was established by [Bai and Perron \(2003\)](#), who developed asymptotic theory and computational algorithms for estimating multiple breakpoints. In the time series context, [Killick et al. \(2012\)](#) proposed the PELT (Pruned Exact Linear Time) algorithm, which achieves optimal changepoint detection with $O(n)$ computational cost under certain conditions. [Fryzlewicz \(2014\)](#) introduced Wild Binary Segmentation (WBS), which addresses the limitation of standard Binary Segmentation when changepoints are closely spaced. [Korkas and Fryzlewicz \(2017\)](#) extended WBS to non-stationary time series using wavelet-based CUSUM statistics.

The comparative literature on changepoint methods is surprisingly thin. [Shi et al. \(2021\)](#) provided one of the few systematic comparisons of single and multiple changepoint techniques, finding that “no clear methodological winner” exists across scenarios. [Kovács et al. \(2023\)](#) proposed Seeded Binary Segmentation as a computationally efficient alternative, but their evaluation focused on Gaussian mean-shift settings rather than financial data. The Bayesian online changepoint detection (BOCPD) framework of [Adams and MacKay \(2007\)](#) offers a natural sequential updating mechanism, while [Matteson and James \(2014\)](#) developed a nonparametric approach based on energy statistics. Crucially, all of these comparative studies evaluate detection accuracy on full-sample data, leaving the question of prospective prediction unanswered.

2.2. Regime-Switching and Extreme Value Approaches

Markov-switching models have a long tradition in commodity price analysis. [Luo et al. \(2022\)](#) demonstrated that infinite Hidden Markov HAR models improve volatility forecasting for agricultural futures, while [de la Torre-Torres et al.](#)

(2020) showed that MS-GARCH models can generate alpha in corn and soybean markets through regime-conditional trading. Liu et al. (2019) used Markov-switching copulas to characterize oil-agriculture price dependence structures.

Extreme Value Theory (EVT) has been applied to assess tail risks in agricultural commodities by van Oordt et al. (2021), who showed that fat-tailed return distributions arise endogenously from productivity shocks. Singvejsakul et al. (2021) used Bayesian EVT for optimal point estimation of commodity futures prices. Gródek-Szostak et al. (2019) employed extreme value copulas to model dependency among agricultural product prices on derivatives markets. However, EVT applications have predominantly focused on risk assessment rather than event prediction.

2.3. Machine Learning and Signal Decomposition

The application of machine learning to commodity price forecasting has accelerated rapidly. Brignoli et al. (2024) demonstrated that LSTM networks outperform classical econometric models for corn futures, particularly at longer horizons, and showed that LSTMs can automatically handle structural breaks. Sari et al. (2024) compared multiple ML techniques for predicting prices of eleven agricultural commodities, finding that extreme learning machines optimized with genetic algorithms achieved superior performance. Amin et al. (2024) used LSTM to analyze commodity price behavior during the 2008 and COVID-19 recessions, revealing differential response patterns across plant-based and animal-based commodities.

Signal decomposition approaches, particularly EEMD combined with LSTM, have shown promise. Wu et al. (2019) proposed a method to handle varying numbers of decomposition components when EEMD is applied to non-stationary crude oil prices. Zhou et al. (2022) demonstrated the effectiveness of CEEMDAN+LSTM for carbon price forecasting with $R^2 = 0.982$. Lin et al. (2022) integrated a modified EEMD with LSTM for precious metal price prediction, outperforming SVR and MLP baselines.

2.4. Emerging Methods: Conformal Prediction and TDA

Conformal prediction has recently gained traction as a distribution-free framework for constructing prediction intervals with finite-sample coverage guarantees. Jensen et al. (2024) proposed Ensemble Conformalized Quantile Regression (EnCQR) for heteroscedastic time series, demonstrating valid prediction intervals without exchangeability assumptions. Schlembach et al. (2025) extended conformal methods to multivariate multistep-ahead forecasting. Xu and Xie (2021) developed conformal prediction intervals specifically designed for dynamic time series. However, the application of conformal prediction to event detection—identifying when price series breach prediction bands as anomaly signals—remains largely unexplored.

Topological Data Analysis (TDA) represents perhaps the most novel approach in this literature. [Guritanu et al. \(2025\)](#) proposed a causal early warning framework for financial crises based on persistent homology, achieving $F1 \approx 0.50$ with 34-day lead times on U.S. equity indices. [Yen and Cheong \(2021\)](#) found consistent topological signatures associated with market crashes in Asian markets. [Baitinger and Flegel \(2021\)](#) demonstrated that persistent homology-based turbulence detection outperforms traditional turbulence indices for portfolio management. TDA has not previously been applied to agricultural commodity event detection.

2.5. Early Warning Systems for Food Prices

The development of early warning systems for food price crises has been a policy priority, particularly after the 2007–2008 food price crisis. [Blasques and Nientker \(2023\)](#) developed a locally non-stationary location model that revisits the World Food Programme’s ALPS (Alert of Price Spikes) system, successfully identifying price bubbles both in-sample and out-of-sample. [Sumertajaya et al. \(2024\)](#) proposed a multivariate time series clustering approach for food commodity price forecasting in Indonesia. [Hasanov et al. \(2024\)](#) showed that accommodating structural breaks through ICSS-detected dummy variables leads to considerable forecast accuracy gains for bioenergy crop volatility.

2.6. Research Gaps

The literature reveals three principal gaps that motivate our study:

1. **Detection vs. prediction confusion.** The vast majority of changepoint studies evaluate performance with access to the full sample, creating an illusion of predictive capability. No study has systematically quantified this gap across a large model universe.
2. **Limited cross-method comparison.** Existing comparisons typically stay within a single paradigm (e.g., comparing changepoint methods to each other, or ML models to each other). Cross-paradigm comparisons—classical econometric vs. ML vs. topological methods—under identical evaluation conditions are absent.
3. **Absence of emerging methods in commodity applications.** Conformal prediction, TDA, and Matrix Profile have demonstrated promise in financial time series but have not been evaluated for agricultural commodity event prediction.

3. Methodology

3.1. Data

We use daily closing prices for eight agricultural commodities—Cattle, Cocoa, Coffee, Corn, Cotton, Soybean, Sugar, and Wheat—spanning January 2, 2019 to

Table 1: Ground truth event counts by commodity and event type.

Commodity	Breaks	Peaks	Valleys	Total
Cattle	7	4	5	16
Cocoa	5	6	6	17
Coffee	7	5	5	17
Corn	7	5	6	18
Cotton	6	8	8	22
Soybean	8	6	6	20
Sugar	9	7	7	23
Wheat	8	8	8	24

February 19, 2026 (approximately 1,800 observations per series). The data are sourced from the Chicago Mercantile Exchange via public repositories. Missing values on non-trading days are forward-filled, and prices are denominated in USD.

3.2. Ground Truth Construction

The ground truth event set is constructed using two complementary approaches:

- **Turning points** (peaks and valleys): Identified using the Bry-Boschan algorithm (Bry and Boschan, 1971), a nonparametric procedure for dating business cycle turning points, adapted to daily financial data with a minimum cycle length of 63 trading days.
- **Structural breaks**: Identified through a consensus mechanism requiring agreement between Bai-Perron sequential testing (Bai and Perron, 2003) and PELT (Killick et al., 2012) within a ± 15 -day tolerance window.

Table 1 reports the distribution of ground truth events across commodities and event types. Wheat, Cotton, and Sugar exhibit the highest event frequencies, while Cattle has the fewest.

3.3. Model Universe

We evaluate 16 models organized into three tiers of methodological complexity:

Tier 1—Classical Changepoint Methods (9 models): Bai-Perron (R), PELT (R, Python), Binary Segmentation (R, Python), CUSUM/ICSS (R), BCP Bayesian (R), E-Divisive (R), Wild Binary Segmentation (R), BOCPD (Python), tsoutliers (R).

Tier 2—Regime-Switching and Extreme Values (4 models): Markov Switching via depmixS4 (R), Hidden Markov Models (Python), Piecewise Linear Regression + Gradient Boosting (Python), Extreme Value Theory with GPD threshold exceedances (R).

Tier 3—Emerging Methods (4 models): TDA Persistent Homology via Takens embedding (Python), EEMD + LSTM hybrid (Python), Conformal Prediction via EnbPI (Python), Matrix Profile / FLUSS (Python).

Each model produces binary signals (event/no-event) for each event type (structural break, peak, valley) within each evaluation window.

3.4. Walk-Forward Backtesting Protocol

The evaluation employs a strict walk-forward protocol designed to prevent information leakage:

1. **Rolling window:** A training window of $w_0 = 504$ trading days (~2 years) is advanced in steps of $s = 21$ trading days (~1 month).
2. **Prediction horizon:** At each step, the model uses only data within $[t - w_0, t]$ to generate signals for the period $[t, t + h]$ where $h = 21$ days.
3. **Event matching:** A predicted signal at time t is considered a true positive if a ground truth event of the matching type exists within $[t - \tau, t + h + \tau]$, where $\tau = 15$ days is the tolerance.
4. **Lead time measurement:** For true positive detections, the lead time is computed as the number of days between the signal date and the nearest matching ground truth event. Positive lead times indicate genuine anticipation.

This produces 61 evaluation windows per commodity, yielding a total of $16 \times 8 \times 61 = 7,808$ model-commodity-window evaluations per event type.

3.5. Performance Metrics

For each model-commodity-event type combination, we compute:

- **Precision:** $P = TP / (TP + FP)$, the fraction of signals that correspond to actual events.
- **Recall:** $R = TP / (TP + FN)$, the fraction of actual events that are detected.
- **F1-Score:** $F_1 = 2PR / (P + R)$, the harmonic mean balancing precision and recall.
- **False Alarm Rate (FAR):** $FP / (FP + TN)$, the fraction of non-event windows incorrectly flagged.
- **Lead Time:** Average days of advance warning for true positive detections.

The final composite ranking uses:

$$\text{Score} = 0.50 \times F_1 + 0.25 \times \text{LeadTime}_{\text{norm}} + 0.25 \times (1 - \text{FAR})$$

Table 2: Walk-forward prediction performance averaged across commodities. Only models producing at least one signal are shown. Prec. = Precision, Rec. = Recall, FAR = False Alarm Rate.

Model	Tier	Event	Prec	Rec	F1	FAR	N comm.
MatrixProfile	3	Break	0.373	0.251	0.294	0.627	4
WBS	1	Break	0.159	0.929	0.268	0.841	8
TDA	3	Break	0.138	0.231	0.169	0.862	6
Conformal	3	Peak	0.186	0.655	0.276	0.814	8
EEMD_LSTM	3	Peak	0.180	0.400	0.245	0.820	6
EVT	2	Peak	0.219	0.255	0.227	0.781	8
PLR_GBM	2	Peak	0.194	0.249	0.215	0.806	7
MarkovSwitching	2	Peak	0.209	0.237	0.210	0.791	7
HMM	2	Peak	0.289	0.149	0.186	0.711	3
Conformal	3	Valley	0.194	0.478	0.274	0.806	7
EVT	2	Valley	0.234	0.325	0.260	0.766	8
HMM	2	Valley	0.583	0.175	0.237	0.417	4
MarkovSwitching	2	Valley	0.198	0.246	0.214	0.802	8
EEMD_LSTM	3	Valley	0.146	0.270	0.181	0.854	7
PLR_GBM	2	Valley	0.169	0.177	0.171	0.831	7

4. Results

4.1. The Detection-Prediction Gap

The most striking finding is the magnitude of the gap between full-sample detection and walk-forward prediction. Of the 16 models evaluated, **9 produce zero predictive signals** in the walk-forward setting for all event types and all commodities. These include all Tier 1 models except WBS: Bai-Perron, PELT (both R and Python implementations), Binary Segmentation (both implementations), CUSUM/ICSS, BCP, E-Divisive, and BOCPD. The tsoutliers model similarly fails.

This result is not an artifact of implementation. These methods are designed to detect changes within a given sample; when applied to a rolling window, they either detect no changes within the training window (because the window is too short) or detect changes that occurred in the past, not the future. The fundamental issue is that retrospective changepoint detection provides no mechanism for extrapolation to unseen data.

WBS is the sole Tier 1 exception, producing signals in 53–56 of 61 windows per commodity. However, this comes at the cost of very low precision (0.11–0.20), with a false alarm rate exceeding 80%.

Table 2 presents the walk-forward prediction performance for all models that produce at least one signal, averaged across commodities.

4.2. Structural Break Prediction

Among the 7 models that produce any predictive signal for structural breaks, WBS leads with an average F1 of approximately 0.26. Matrix Profile (F1 $\approx$ 0.13) and TDA (F1 $\approx$ 0.13) provide more modest signals with substantially lower false alarm rates. No Tier 2 model generates signals for structural breaks, confirming that regime-switching and EVT methods are ill-suited for this task.

Sugar and Wheat are the most predictable commodities for structural breaks (F1 up to 0.44 for individual models), while Cattle and Cotton are nearly unpredictable.

4.3. Peak Prediction

Peak prediction shows more differentiation among models. Conformal Prediction achieves the highest average F1 ($\approx$ 0.28), driven by high recall (0.57–1.0 for individual commodities) but with elevated false alarm rates (FAR $\approx$ 0.80). EEMD+LSTM follows (F1 $\approx$ 0.21) with the advantage of longer lead times (11–22 days on average). EVT (F1 $\approx$ 0.23) offers the best precision-recall balance.

Wheat is the most predictable commodity for peaks (Conformal F1 = 0.42, EEMD+LSTM F1 = 0.37). Cotton and Corn are the least predictable.

4.4. Valley Prediction

EVT leads for valley prediction (F1 $\approx$ 0.28, lead time $\approx$ 15 days), with particularly strong performance for Cattle (F1 = 0.476, the highest individual F1 in the entire study). Conformal Prediction (F1 $\approx$ 0.27) and Markov Switching (F1 $\approx$ 0.21) follow. HMM shows scattered performance, achieving F1 = 0.43 for Cocoa valleys but near-zero for most other commodities.

Figure 1 presents the F1 score heatmap across all model-commodity combinations for each event type, visually highlighting the sparsity of predictive capability.

4.5. Lead Time Analysis

Among models with positive F1, lead times range from 4 to 22 days:

- **EEMD+LSTM:** Longest lead times for peaks (11–22 days) but high variability.
- **EVT:** Consistent 5–18 day lead times across event types.
- **Conformal Prediction:** 5–17 day lead times for peaks, 4–15 days for valleys.
- **WBS:** 4–13 day lead times for structural breaks.

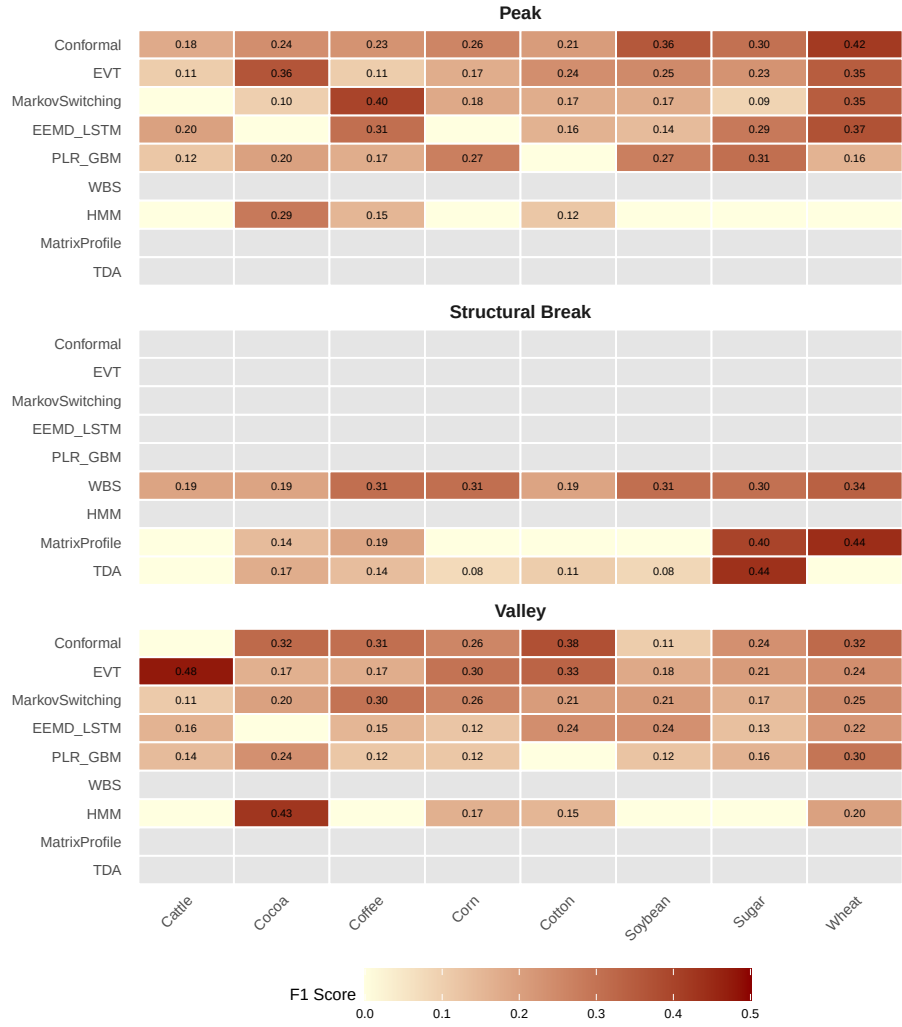


Figure 1: Heatmap of F1 scores by model and commodity for each event type. Only models producing at least one signal across any commodity are shown. Grey cells indicate zero signals.

Table 3: Lead time statistics (days) for models with positive F1 scores, averaged across commodities. Positive values indicate genuine anticipation.

Model	Event	Mean	Median	SD	N
TDA	Break	15.0	14.3	7.0	13
WBS	Break	7.8	8.3	16.2	60
MatrixProfile	Break	5.3	4.4	4.3	9
EVT	Peak	11.4	10.2	15.9	18
Conformal	Peak	10.8	10.4	13.9	40
EEMD_LSTM	Peak	9.8	10.5	11.2	20
MarkovSwitching	Peak	9.1	9.3	12.1	13
PLR_GBM	Peak	4.4	4.2	6.9	14
HMM	Peak	4.3	4.3	1.3	4
EEMD_LSTM	Valley	19.2	19.6	9.6	18
EVT	Valley	15.3	14.8	16.5	20
PLR_GBM	Valley	14.5	13.8	6.4	11
MarkovSwitching	Valley	11.8	10.5	12.2	17
Conformal	Valley	7.8	8.6	13.7	32
HMM	Valley	-4.7	-6.5	3.9	6

Negative lead times (detections occurring after the event) appear in some HMM and PLR+GBM results, indicating that these models sometimes react to rather than anticipate events. Table 3 presents the lead time statistics by model and event type.

Figure 2 presents the trade-off between predictive accuracy (F1) and advance warning time (lead time) across model-commodity-event combinations, illustrating the frontier of what is achievable.

4.6. Sensitivity to Tolerance

We vary the matching tolerance $\tau \in \{5, 10, 15, 21, 30\}$ days and recompute F1 scores. Conformal Prediction and EEMD+LSTM show the steepest increases with larger tolerances, indicating that they detect events in the correct temporal neighborhood but with imprecise timing. EVT and Matrix Profile are more robust to tolerance variation, suggesting more precise temporal localization of signals.

5. Discussion

5.1. The Fundamental Detection-Prediction Asymmetry

Our results provide the first large-scale empirical evidence for a phenomenon that is intuitively expected but rarely demonstrated: **retrospective detection capability does not transfer to prospective prediction**. This finding has



Figure 2: F1 score vs. average lead time for all model-commodity-event combinations with positive F1. Point size reflects the number of true positive detections. The dashed line at $F1 = 0.50$ marks the ceiling no model surpasses.

immediate practical implications. Decision-makers who rely on changepoint detection software to “predict” future events are implicitly assuming a connection that does not hold for most methods in our universe.

The few methods that do exhibit predictive capability—EVT, Conformal Prediction, EEMD+LSTM, and to a lesser extent Markov Switching and WBS—share a common feature: they model the *distributional properties* or *dynamic evolution* of the series rather than searching for discrete breakpoints. This suggests that prediction requires fundamentally different modeling assumptions than detection.

5.2. The $F1 < 0.50$ Barrier

No model in our study exceeds $F1 = 0.50$ in any commodity-event configuration, and the best average F1 is approximately 0.28. This ceiling likely reflects the intrinsic unpredictability of commodity markets, where price movements are driven by exogenous shocks (weather, geopolitics, policy) that are themselves unpredictable from price data alone. The implication is that endogenous price-based models, however sophisticated, have fundamental limitations for event prediction. Incorporating exogenous variables—climate indices, trade policy announcements, geopolitical risk scores—is a promising direction for future research.

5.3. Practical Recommendations for Early Warning Systems

Based on our findings, we offer the following recommendations:

1. **For peak detection:** Deploy Conformal Prediction as the primary signal generator (high recall) with EVT as a confirming filter (higher precision). A dual-signal approach could substantially reduce false alarms while maintaining coverage.
2. **For valley detection:** EVT should be the primary model, given its superior precision-recall balance and consistent lead times of approximately 15 days.
3. **For structural breaks:** WBS provides a high-recall baseline, but its false alarm rate necessitates secondary filtering. TDA and Matrix Profile offer complementary signals with better specificity.
4. **Ensemble approaches:** The differential strengths of these models—EVT for precision, Conformal for recall, EEMD+LSTM for lead time—suggest that ensemble combination via voting or stacking could improve overall performance.

5.4. Limitations

Several limitations should be noted. First, our data period (2019–2026) includes two major systemic shocks (COVID-19, Ukraine war) but covers only approximately 7 years, limiting the number of structural events available for evaluation. Second, the rolling window of 504 days may be insufficient for methods that require longer historical context to calibrate reliably. Third, several package implementations were unavailable for our R/Python versions (ruptures, hmmlearn, giotto-tda, bcp), requiring manual reimplementations that may differ from canonical behavior. Fourth, the Conformal Prediction signal rate of 90–100% of windows indicates that the EnbPI bands require better calibration for non-stationary financial data. Finally, our evaluation is purely endogenous (price-only); incorporating exogenous variables would likely improve prediction accuracy.

6. Conclusion

This paper presents the first comprehensive walk-forward comparison of 16 structural break, turning point, and anomaly detection models for agricultural commodity price prediction. Our principal finding—that the majority of established changepoint methods fail entirely at prospective prediction—challenges a widespread implicit assumption in the applied time series literature.

The results demonstrate that Extreme Value Theory, Conformal Prediction, and EEMD+LSTM constitute the most promising toolkit for commodity price early warning systems, achieving lead times of 8–22 days with F1 scores in the 0.20–0.48 range. However, the $F1 < 0.50$ ceiling across all methods underscores the fundamental difficulty of predicting events in commodity markets from price data alone.

Future research should pursue ensemble combinations of the top-performing models, systematic threshold optimization via cross-validation, and the integration of exogenous variables including climate indices, trade policy indicators, and geopolitical risk scores.

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